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Substrate susceptor using edge purging

Abstract

A workpiece susceptor body can include a front face configured to support a workpiece, a back face opposite the front face, a workpiece contact zone at least partially forming a support boundary on an inner portion of the front face, and a plurality of axial channels disposed within the susceptor body. The workpiece contact zone can be disposed radially inward of an outer edge of a workpiece positioned on the front face in a processing configuration. Each of the plurality of axial channels may connect to corresponding openings extending into an outer portion of the front face. Each of the openings may be disposed radially outward of the workpiece contact zone of the susceptor body.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3407783	12/1967	Capita	N/A	N/A
3549847	12/1969	Clark et al.	N/A	N/A
3641974	12/1971	Yamada et al.	N/A	N/A
3796182	12/1973	Rosler	N/A	N/A
4499354	12/1984	Hill et al.	N/A	N/A
4522149	12/1984	Garbis et al.	N/A	N/A
4560420	12/1984	Lord	N/A	N/A
4710428	12/1986	Tamamizu et al.	N/A	N/A
4821674	12/1988	deBoer et al.	N/A	N/A
4958061	12/1989	Wakabayashi et al.	N/A	N/A
4978567	12/1989	Miller	N/A	N/A
4986215	12/1990	Yamada et al.	N/A	N/A
4990374	12/1990	Keeley et al.	N/A	N/A
5033407	12/1990	Mizuno et al.	N/A	N/A
5044943	12/1990	Bowman et al.	N/A	N/A
5071670	12/1990	Kelly	N/A	N/A
5074017	12/1990	Toya et al.	N/A	N/A
5098198	12/1991	Nulman et al.	N/A	N/A
5108792	12/1991	Anderson et al.	N/A	N/A
5119540	12/1991	Kong et al.	N/A	N/A
5119541	12/1991	Ohmi et al.	N/A	N/A
5121531	12/1991	Severns et al.	N/A	N/A
5156820	12/1991	Wong et al.	N/A	N/A
5188501	12/1992	Tomita et al.	N/A	N/A
5199483	12/1992	Bahng	N/A	N/A

5200157	12/1992	Toya et al.	N/A	N/A
5242501	12/1992	McDiarmid	N/A	N/A
5292554	12/1993	Sinha et al.	N/A	N/A
5298465	12/1993	Levy	N/A	N/A
5304248	12/1993	Cheng et al.	N/A	N/A
5306699	12/1993	Eddy	N/A	N/A
5308645	12/1993	Zachman et al.	N/A	N/A
5332442	12/1993	Kubodera et al.	N/A	N/A
5343938	12/1993	Schmidt	N/A	N/A
5354715	12/1993	Wang et al.	N/A	N/A
5356486	12/1993	Sugarman et al.	N/A	N/A
5370739	12/1993	Foster et al.	N/A	N/A
5383971	12/1994	Selbrede	N/A	N/A
5393349	12/1994	Ohkase	N/A	N/A
5403401	12/1994	Haafkens et al.	N/A	N/A
5421893	12/1994	Perlov	N/A	N/A
5427620	12/1994	deBoer et al.	N/A	N/A
5444217	12/1994	Moore et al.	N/A	N/A
5455069	12/1994	Lee	N/A	N/A
5456757	12/1994	Aruga et al.	N/A	N/A
5461214	12/1994	Peck et al.	N/A	N/A
5467220	12/1994	Xu	N/A	N/A
5492566	12/1995	Sumnitsch	N/A	N/A
5514439	12/1995	Sibley	N/A	N/A
5516367	12/1995	Lei et al.	N/A	N/A
5527393	12/1995	Sato et al.	N/A	N/A
5549756	12/1995	Sorensen et al.	N/A	N/A
5551985	12/1995	Brors et al.	N/A	N/A
5558717	12/1995	Zhao et al.	N/A	N/A
5584936	12/1995	Pickering et al.	N/A	N/A
5588827	12/1995	Muka	N/A	N/A
5620525	12/1996	van de Ven et al.	N/A	N/A
5645646	12/1996	Beinglass et al.	N/A	N/A
5656093	12/1996	Burkhart et al.	N/A	N/A
5683518	12/1996	Moore et al.	N/A	N/A
5690742	12/1996	Ogata et al.	N/A	N/A
5700725	12/1996	Hower et al.	N/A	N/A
5738165	12/1997	Imai	N/A	N/A
5761023	12/1997	Lue et al.	N/A	N/A
5792304	12/1997	Tamura et al.	N/A	N/A
5800622	12/1997	Takemi et al.	N/A	N/A
5803977	12/1997	Tepman et al.	N/A	N/A
5834737	12/1997	Hirose et al.	N/A	N/A
D404370	12/1998	Kimura	N/A	N/A
5895530	12/1998	Shrotriya et al.	N/A	N/A
5938850	12/1998	Arami et al.	N/A	N/A
5960159	12/1998	Ikeda et al.	N/A	N/A
6001183	12/1998	Gurary et al.	N/A	N/A
6020212	12/1999	Mathis	N/A	N/A

6040011	12/1999	Yudovsky	118/728	H01L 21/68792
6068441	12/1999	Raaijmakers et al.	N/A	N/A
6077357	12/1999	Rossman et al.	N/A	N/A
6086680	12/1999	Foster et al.	N/A	N/A
6090212	12/1999	Mahawili	N/A	N/A
6093252	12/1999	Wengert et al.	N/A	N/A
6113702	12/1999	Halpin	118/728	H01L 21/68735
6167834	12/2000	Wang et al.	N/A	N/A
6293749	12/2000	Raaijmakers et al.	N/A	N/A
6325858	12/2000	Wengert et al.	N/A	N/A
6331023	12/2000	Goodwin et al.	N/A	N/A
6343183	12/2001	Halpin et al.	N/A	N/A
6394797	12/2001	Sugaya et al.	N/A	N/A
6402850	12/2001	Beinglass et al.	N/A	N/A
6464790	12/2001	Sherstinsky et al.	N/A	N/A
6608287	12/2002	Halpin et al.	N/A	N/A
6634882	12/2002	Goodman	N/A	N/A
6709267	12/2003	Hawkins et al.	N/A	N/A
D496008	12/2003	Takahashi et al.	N/A	N/A
6893507	12/2004	Goodman et al.	N/A	N/A
7033445	12/2005	Keeton et al.	N/A	N/A
D525127	12/2005	Cogley et al.	N/A	N/A
D600223	12/2008	Aggarwal et al.	N/A	N/A
7601224	12/2008	Foree	N/A	N/A
7602224	12/2008	Foree	N/A	N/A
7648579	12/2009	Goodman et al.	N/A	N/A
7722441	12/2009	Goela et al.	N/A	N/A
7740703	12/2009	Hasegawa et al.	N/A	N/A
8088225	12/2011	Goodman et al.	N/A	N/A
8287648	12/2011	Reed et al.	N/A	N/A
8366830	12/2012	Nakamura et al.	N/A	N/A
8394229	12/2012	Aggarwal et al.	N/A	N/A
8801857	12/2013	Aggarwal et al.	N/A	N/A
D743357	12/2014	Vyne	N/A	N/A
9558985	12/2016	Wang et al.	N/A	N/A
D784276	12/2016	Tiner et al.	N/A	N/A
9633889	12/2016	Cuvalci et al.	N/A	N/A
9698042	12/2016	Baldasseroni et al.	N/A	N/A
9719169	12/2016	Mohn et al.	N/A	N/A
10068791	12/2017	Parsey, Jr. et al.	N/A	N/A
D830981	12/2017	Jeong et al.	N/A	N/A
10242848	12/2018	Jeon et al.	N/A	N/A
10242890	12/2018	Volfovski et al.	N/A	N/A
10376916	12/2018	Ranish et al.	N/A	N/A
10388557	12/2018	Kawamura et al.	N/A	N/A

D864134	12/2018	Watarai et al.	N/A	N/A
D914620	12/2020	Rokkam et al.	N/A	N/A
11018047	12/2020	Rokkam et al.	N/A	N/A
D920936	12/2020	Rokkam et al.	N/A	N/A
2002/0011211	12/2001	Halpin	N/A	N/A
2002/0043337	12/2001	Goodman et al.	N/A	N/A
2002/0144786	12/2001	Chiang	156/345.52	C23C 16/45561
2003/0000472	12/2002	Lim et al.	N/A	N/A
2003/0049580	12/2002	Goodman	N/A	N/A
2003/0075109	12/2002	Arai	118/728	C23C 16/4581
2004/0060512	12/2003	Waldhauer et al.	N/A	N/A
2004/0229002	12/2003	Davis et al.	N/A	N/A
2005/0092439	12/2004	Keeton et al.	N/A	N/A
2005/0193952	12/2004	Goodman	118/728	H01L 21/68735
2005/0284372	12/2004	Murugesh et al.	N/A	N/A
2006/0057826	12/2005	De Boer	N/A	N/A
2009/0235866	12/2008	Kataigi et al.	N/A	N/A
2009/0280248	12/2008	Goodman et al.	N/A	N/A
2010/0031884	12/2009	Aggarwal et al.	N/A	N/A
2010/0107974	12/2009	Givens et al.	N/A	N/A
2012/0211933	12/2011	Goto	N/A	N/A
2013/0109192	12/2012	Hawkins et al.	N/A	N/A
2016/0002778	12/2015	Ravi et al.	N/A	N/A
2016/0068996	12/2015	Lau et al.	N/A	N/A
2016/0133504	12/2015	Chu et al.	N/A	N/A
2017/0175265	12/2016	Shah	N/A	C23C 16/4584
2018/0094350	12/2017	Verghese et al.	N/A	N/A
2018/0138074	12/2017	Lee et al.	N/A	N/A
2019/0139807	12/2018	White	N/A	H01L 21/68764
2019/0355612	12/2018	Sakurai	N/A	N/A
2020/0234996	12/2019	Rokkam et al.	N/A	N/A
2020/0373187	12/2019	Singu	N/A	B25B 11/005
2021/0125853	12/2020	Rathi et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102308362	12/2011	CN	N/A
103733327	12/2013	CN	N/A
106463365	12/2016	CN	N/A
0339279	12/1988	EP	N/A
0448346	12/1990	EP	N/A
0445596	12/1990	EP	N/A
0634785	12/1994	EP	N/A
0669640	12/1994	EP	N/A

0766289	12/1996	EP	N/A
0840358	12/1997	EP	N/A
2181458	12/1986	GB	N/A
7-58039	12/1994	JP	N/A
2000-269310	12/1999	JP	N/A
2002-184843	12/2001	JP	N/A
2002-526915	12/2001	JP	N/A
2003-124167	12/2002	JP	N/A
2006-228802	12/2005	JP	N/A
2007-502022	12/2006	JP	N/A
D1422195	12/2013	JP	N/A
D1547357	12/2015	JP	N/A
D1570747	12/2017	JP	N/A
D1570748	12/2017	JP	N/A
10-1999-0069084	12/1998	KR	N/A
1999-0069091	12/1998	KR	N/A
100460338	12/2004	KR	N/A
10-2007-0098025	12/2006	KR	N/A
D111900	12/2005	TW	N/A
D136587	12/2009	TW	N/A
D184695	12/2016	TW	N/A
D191199	12/2017	TW	N/A
WO 1996/30713	12/1995	WO	N/A
WO 1997/08743	12/1996	WO	N/A
WO 2010/016964	12/2009	WO	N/A

OTHER PUBLICATIONS

“Shaping the Future in Semiconductor Processing.” Advertisement for Morton International, Inc., Ceramic Solutions. vol. 73, No. 7, Jul. 1994: advertisements (circa 1993). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. patent application Ser. No. 16/875,088, filed May 15, 2020, which claims the benefit of priority to U.S. Provisional Application No. 62/851,414, filed May 22, 2019, which is hereby incorporated by reference herein in its entirety. Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

Field of the Disclosure

(1) This disclosure relates generally to semiconductor processing, and more particularly to susceptors for supporting semiconductor substrates in process chambers.

Description of Related Art

(2) Semiconductor fabrication processes are typically conducted with the substrates supported

within a reaction chamber on a susceptor under controlled process conditions. For many processes, semiconductor substrates (e.g., wafers) are heated inside the reaction chamber. A number of quality control issues related to the physical interaction between the substrate and the susceptor can arise during processing.

SUMMARY

(3) In some embodiments, a workpiece susceptor is provided. The workpiece susceptor body comprises a front face configured to support a workpiece and a back face opposite the front face. The workpiece susceptor also includes a workpiece contact zone at least partially forming a support boundary around an inner portion of the front face. The workpiece contact zone is configured to be disposed radially inward of an outer edge of a workpiece positioned on the front face in a processing configuration. The workpiece susceptor also includes one or more axial channels that are disposed within the susceptor body. The axial channels connect to one or more openings extending into an outer portion of the front face. Each of the openings are disposed radially outward of the workpiece contact zone of the susceptor body. The workpiece contact zone is at a greater elevation than the outer portion of the face to form a gap radially outward of the workpiece contact zone and axially between the face of the susceptor body and the workpiece.

(4) In some embodiments, a method for purging a workpiece susceptor is provided. The method comprises loading a workpiece onto a workpiece contact zone on a front face of a susceptor body, such that an outer edge of the workpiece is disposed radially outward from the workpiece contact zone. The method further includes positioning the workpiece in a processing configuration such that the front face of the susceptor body is in fluid communication with a reaction chamber and such that a back face of the susceptor body is in fluid communication with a loading chamber. The method includes providing a first pressure within the reaction chamber and purging a backside of the outer edge of the workpiece by flowing purge gas at a second pressure from one or more channels within the susceptor body, to a gap radially outward of the workpiece contact zone and axially between the front face of the susceptor plate and the workpiece, to the reaction chamber. The second pressure is greater than the first pressure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above, as well as additional objects, features and advantages of the present inventive concept, will be better understood through the following illustrative and non-limiting detailed description of embodiments of the present invention, with reference to the appended drawings. In the drawings like reference numerals will be used for like elements unless stated otherwise.
- (2) FIG. 1 schematically illustrates a cross-section of a semiconductor processing apparatus with a workpiece support in a loading position, according to one embodiment.
- (3) FIG. 2 schematically illustrates the apparatus of FIG. 1 with the workpiece support shown in a processing position, according to one embodiment.
- (4) FIG. 3 shows an example susceptor body that includes a pedestal that supports a back face and a front face.
- (5) FIG. 4A shows a schematic cross-sectional side view of a portion of the susceptor body.
- (6) FIG. 4B shows a perspective cross-sectional view of an example susceptor body with a pedestal.
- (7) FIG. 4C shows an example susceptor body that includes axial purge channels that connect a front face of the body with a back face of the body.
- (8) FIG. 5 shows a cross-sectional view of an example susceptor body.
- (9) FIG. 6 shows a perspective of a cross-section of an example susceptor body.
- (10) FIG. 7 shows a cross-sectional top view of an example susceptor body sectioned between the

front face and the back face.

(11) FIG. 8 shows an example method for purging a workpiece susceptor.

DETAILED DESCRIPTION

(12) Although certain embodiments and examples are disclosed below, it will be understood by those having ordinary skill in the art that the invention extends beyond the specifically disclosed embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Thus, it is intended that the scope of the invention herein disclosed should not be limited by the particular disclosed embodiments described below.

(13) Substrate preparation and processing can require precision manufacturing techniques and apparatus. Moreover, their processing can require maintenance and cleaning. Described herein are various embodiments for improving the cleanliness and quality of substrate processing.

(14) Various types of substantially flat workpieces, such as substrates (e.g., wafers, including semiconductor wafers) are commonly processed on susceptor assemblies within a processing apparatus. Susceptor assemblies may include a susceptor, which can be formed by machining various hard materials (e.g., ceramic, graphite) into a desired shape and optionally applying a coating (e.g., silicon carbide (SiC)). Susceptors can be formed in different shapes to support workpieces of various shapes, but many are circular.

(15) As noted above, a number of quality control issues can arise during processing, relating to the interaction between the reactant and the susceptor, particularly during processing of the substrate. These issues can include, for example, backside deposition on the substrate, undesired deposition on various parts or regions of the susceptor assembly and/or processing apparatus. Such quality control issues can decrease the overall quality and production of the substrates and semiconductor devices, resulting in reduced yield and increased costs.

(16) Backside deposition occurs when process gases flow into the space between the substrate and the susceptor and deposit on a back surface of the substrate. Because the flow of the process gases is not controlled between the substrate and the susceptor, random deposition can occur on the backside of the substrate. In addition to issues noted above, this random deposition can create thickness inconsistencies on the backside, which can affect local site flatness on the frontside, and ultimately cause device uniformity issues.

(17) In a typical process, a reactant gas is passed over the heated wafer, causing the chemical vapor deposition (CVD) of a thin layer of reactant material on the wafer. Through sequential processing, multiple layers are made into integrated circuits. Other exemplary processes include sputter deposition, photolithography, dry etching, plasma processing, and high temperature annealing. Many of these processes require high temperatures and can be performed in the same or similar reaction chambers. The reactant gas can include a metal chloride (e.g., titanium chloride, titanium tetrachloride, etc.), other metal compounds (e.g., molybdenum, tungsten, etc.), a silicon-based gas (e.g., silane, di-silane, tri-silane, etc.), an oxidation gas (e.g., H_2O , O_2 , O_3 , etc.), and/or any other suitable reactant gas. Various films can be prepared, such as metal-based (e.g., titanium, molybdenum, tungsten, etc.) films, silicon-based films, and/or other films. The film may include an oxide layer. In some embodiments, the reactant gas can include a mixture of any of the above gases (or combinations thereof) with other gases, such as hydrogen. In some embodiments, two or more reactant gases can be implemented. For example, a first reactant gas (e.g., molybdenum gas) may be used which excludes hydrogen and/or a second reactant gas may include a mixture with hydrogen (e.g., molybdenum gas plus hydrogen). The second reactant gas may be passed over the substrate after the first reactant gas.

(18) Wafers may be processed at various temperatures to promote high quality deposition.

Temperature control is especially helpful at temperatures below the mass transport regime, such as about 500° C. to 900° C. for silicon CVD using silane. In this kinetic regime, if the temperature is not uniform across the surface of the wafer, the deposited film thickness will be uneven. However, lower temperatures may sometimes be used in certain scenarios.

(19) The diameter of the wafer can influence the processing as well. In recent years, single-wafer processing of large diameter wafers has become more widely used for a variety of reasons, including the desire for greater precision in process control than may be achieved with batch-processing. Wafers may be made of silicon, most commonly with a diameter of about 150 mm (about 6 inches) or of about 200 mm (about 8 inches) and with a thickness of about 0.725 mm. Recently, larger silicon wafers with a diameter of about 300 mm (about 12 inches) and a thickness of about 0.775 mm have been utilized because they exploit the benefits of single-wafer processing even more efficiently. Even larger wafers are expected in the future. A typical single-wafer susceptor includes a pocket or recess within which the wafer rests during processing. In many cases, the recess is shaped to receive the wafer very closely.

(20) To address some of the issues described above, a susceptor assembly or processing apparatus can include purge channel designs. As noted, these improvements may reduce reactant within a reaction chamber from coming in contact with a backside of a substrate (e.g., wafer), a face of a susceptor assembly, or other undesired areas. When reactant inadvertently comes in contact with certain areas of the processing apparatus or susceptor assembly, the corresponding areas may need to be cleaned. This can create not only additional labor in processing substrates but may also reduce the quality of future wafer processing.

(21) Reference will now be made to the Figures.

(22) FIG. 1 schematically illustrates an embodiment of a semiconductor processing apparatus **100** comprising a reaction chamber **101** and a loading chamber **102**. Together, the reaction chamber **101** and the loading chamber **102** may be considered a process module, for example, to be implemented into multi-module “cluster” tools. In the illustrated embodiment, the reaction chamber **101** is disposed above the loading chamber **102**, and they are separated by a baseplate **107** and a movable pedestal or workpiece support **109**, described in more detail below.

(23) In some embodiments, the reaction chamber **101** may be substantially smaller than the loading chamber **102**, contrary to the schematic drawings, which are not drawn to scale. For a single wafer process module, as shown, the reaction chamber **101** may have a volume between about 0.25 liters and 3 liters. In some embodiments, the reaction chamber **101** may have a volume of less than about 1 liter. In some embodiments, the reaction chamber **101** may be about 900 mm long, 600 mm wide, and 5 mm high. In some embodiments, the loading chamber **102** may have a volume between about 30 liters and about 50 liters. In some embodiments, the loading chamber **102** may have a volume of about 40 liters. In some embodiments, the loading chamber **102** may have a volume about 35-45 times the volume of the reaction chamber **101**.

(24) In some embodiments, the reaction chamber **101** may comprise one or more inlets **103** (one shown) and one or more outlets **104** (one shown). During processing, gases such as reactants and purge gases may flow into the reaction chamber **101** through the reaction chamber inlet **103**, and gases such as excess reactants, reactant byproducts, and purge gases may flow out of the reaction chamber **101** through the reaction chamber outlet **104**. In some embodiments, the loading chamber **102** may comprise one or more inlets **105** (one shown) and one or more outlets **106** (one shown). In operation, gases such as purge gases may flow into the loading chamber **102** through the loading chamber inlet **105**, and gases such as excess reactants, reactant byproducts, and purge gases may flow out of the loading chamber **102** through the loading chamber outlet **106**. The depicted configuration, such as the positions of the inlets **103**, **105** and outlets **104**, **106** are merely schematic, and may be adjusted based on, for example, the process to be performed in the reaction chamber **101**, the desired flow path of the gases, etc. Purge gases can include a single purge gas or a mixture of purge gases. For example, in some embodiments, the purge gas can consist essentially of one or more inert gases, such as one or more noble gases (e.g., helium, argon, neon, xenon, etc.). The purge gas can include one or more inert gases without any reactive gases. In other embodiments, the purge gas can include, for example, one or more inert gases and one or more other non-inert gases. The purge gas can include an inert gas mixed with a reactive gas, such as

hydrogen. The purge gas may include a mixture of hydrogen and argon, for example. In some embodiments, a first purge gas consisting essentially of one or more inert gases (i.e., without any reactive gases) can be used in a first purge step, and a second purge gas comprising a mixture of one or more inert gases mixed with one or more reactive gases can be used in a second purge step. In some embodiments, this second purge step sequentially follows this first purge step. Using a purge step that includes one or more inert gases with one or more reactive gases may help improve the distribution of a reactant across the substrate. For example, a delivery system (e.g., shower) may generally concentrate the reactant near a center of the substrate. During a second purge step, a mixture of inert and reactive gases can provide a better distribution of reactant near, for example, the edges of the substrate.

(25) In the illustrated embodiment, the reaction chamber **101** comprises a baseplate **107** including an opening **108**. An interior edge of the baseplate **107** defines the opening **108**. In some embodiments, the baseplate **107** may comprise titanium. In the illustrated embodiment, the reaction chamber inlet **103** is located approximately opposite to the reaction chamber outlet **104**, such that reaction gas that flows from the reaction chamber inlet **103** to the reaction chamber outlet **104** travels approximately parallel to a face of the workpiece W, and thus parallel to the upper surface of the moveable support. Such reactors are sometimes referred to as “cross-flow” or horizontal laminar flow reactors. In some embodiments, the apparatus **100** may be an atomic layer deposition (ALD) reactor, such that it includes valves controlled by a control system **113** to separately provide pulses of reactants. In some embodiments, apparatus **100** may include two or more valves independently controlled by control system **113** to allow regulation of relative pressure and/or the direction of flow between reaction chamber **101** and loading chamber **102**. In some embodiments, the reaction chamber inlet **103** may comprise a distribution system such to distribute gas in a desirable pattern. In some embodiments, the reaction chamber **101** may taper near the reaction chamber outlet **104**, such that the height of the reaction chamber **101** decreases near the reaction chamber outlet **104**, thereby constricting air flow through the reaction chamber outlet **104**. Although the apparatus **100** may be described herein with respect to vapor deposition (e.g., chemical vapor deposition, or CVD, and/or atomic layer vapor deposition, or ALD) reactors, the apparatus **100** may alternatively comprise other semiconductor processing tools, including, but not limited to, dry etchers, ashers, rapid thermal annealers, etc.

(26) The apparatus **100** further comprises the moveable support **109**, configured to be moved between a loading position and a processing position by operation of a drive mechanism **110**. FIG. 1 depicts the support **109** in the loading position, according to one embodiment. The support **109** may be configured to hold a workpiece (semiconductor workpiece W see FIG. 2), such as a silicon wafer. The workpiece W may be loaded and unloaded into the support **109** in various ways, such as with an end effector of a robot. The support **109** may comprise lift-pins **111** and/or cutouts to aid in loading and unloading of the workpiece W with a paddle or fork. The support **109** may comprise a vacuum system that holds the workpiece W in place after loading, or gravity alone may hold the workpiece W in a pocket that is sized and shaped to accommodate the workpiece W. The apparatus **100** may further comprise one or more gate valves **112** (one shown) for loading and unloading of workpieces W to and from the support **109**. The gate valve **112** may allow access to, for example, a transfer chamber, load lock, processing chamber, clean room, etc.

(27) The control system **113** is also configured or programmed to control the drive mechanism **110**. In some embodiments, the drive mechanism **110** may comprise a piston or elevator that imparts vertical movement to the support **109**. The drive mechanism **110** is therefore configured to move the support **109**, and thus the workpiece W disposed on the support **109**, into the processing position during a reactor closure operation and into the loading position during a reactor opening operation. The drive mechanism **110** can also be configured to rotate the workpiece W disposed on the support **109**.

(28) FIG. 2 schematically illustrates the apparatus **100** with the support **109** shown in the

processing position, according to one embodiment. When in the processing position, the support **109** engages the baseplate **107**, effectively isolating or separating the interior of the reaction chamber **101** from the loading chamber **102**. Such isolation can reduce contamination between the reaction chamber **101** and the loading chamber **102**. In some embodiments, engaging may comprise creating a hard metal-on-metal seal between the baseplate **107** and the support **109**. In some embodiments, engaging may comprise compression of pliable material, such as an O-ring, on either part, to create a soft seal between the baseplate **107** and the support **109**. In some embodiments, engaging may comprise maintaining a gap between the support **109** and the baseplate **107**, such that there is no absolute seal. Even where engaging comprises maintaining a gap between the support **109** and the baseplate **107**, the support may still effectively separate the reaction chamber **101** from the loading chamber **102** by creating a substantial barrier to fluid communication between the reaction chamber **101** and the loading chamber **102** when apparatus **100** is in the processing position.

(29) FIG. 3 shows an example susceptor body **200**. The support **109** illustrated in FIGS. 1 and 2 can include the susceptor body **200**. The susceptor body **200** can include a back face **204** and a front face **202**. The susceptor body **200** can include a pedestal **218** that supports the back face **204** and the front face **202**. The front face **202** can be opposite the back face **204**. The front face **202** may be substantially flat throughout, with some features extending into or from the front face **202** as described herein. The front face **202** can be configured to support a workpiece, such as a substrate (e.g., a wafer). The front face **202** and/or the back face **204** may form one or more shapes, such as a circle, ellipse, rectangle, etc. As shown in FIG. 3, the front face **202** may be substantially circular. Accordingly, the term “circumferential” or related terms are used throughout. However, the concepts described herein may be applied for other shapes as well.

(30) The body **200** can include one or more openings **222**. The opening(s) **222** can extend into an outer portion **220** of the front face **202**. The openings **222** can provide edge purging to a workpiece supported on the front face **202**, as described further herein. The openings **222** can be in fluid communication with one or more axial channels (e.g., channel(s) **252** shown in FIGS. 4A-4C) disposed within the susceptor body **200**, for example, at least partially between the front face **202** and the back face **204**. In some embodiments, the body can include a circumferentially extending axial channel. For example, a circumferentially extending axial channel **224** can extend into the front face **202**. The opening(s) **222** can be in communication with the circumferentially extending axial channel **224**, as shown. The circumferentially extending axial channel can extend in a complete circumference, to form a ring-shaped channel, as shown.

(31) The purge openings **222** may be configured to have a fixed or variable cross sectional area, to control flow there through. For example, the purge openings **222** may include a corresponding flow control valve, such as a needle valve, to provide adjustable flow. In some embodiments, the openings **222** can comprise fixed orifices with a desired cross sectional area (e.g., corresponding to a given diameter) and/or in desired quantities to provide a desired amount of edge purge flow under some process conditions. For example, the susceptor body **200** can have between about 15 and 36 purge openings **222** and in some embodiments has 18. Each of the purge openings **222** can have a diameter in a range between about 0.5 mm and 2 mm, and in some embodiments the diameter is about 1.85 mm.

(32) The susceptor body **200** can include a workpiece contact zone **210**. The workpiece contact zone **210** may at least partially form a support boundary around an inner portion **226** of the front face **202**. In some embodiments, a seal can be formed at the interface between the workpiece contact zone **210** and a substrate held thereupon, to prevent flow between the outer portion **220** and inner portion **226** of the front face **202**. The workpiece contact zone **210** can be configured to be disposed radially inward of the outer edge of a workpiece that is positioned on the front face **202**. The workpiece contact zone **210** can be at a greater elevation than the outer portion **220** of the front face **202** to form a gap (not shown) radially outward of the workpiece contact zone **210** and axially

between the front face **202** of the susceptor body **200** and the workpiece when the workpiece is positioned on the front face **202**. The workpiece contact zone **210** can include a circumferential rib extending from the front face **202**. Each of the openings **222** can be disposed radially outward of the workpiece contact zone **210**. The workpiece contact zone **210** can have a thickness (i.e., radial width) in a range between about 1.5 mm and about 3 mm, and in some embodiments the thickness is about 2 mm. In embodiments where it is circular, the workpiece contact zone **210** can form an approximately annular support ring having a diameter in a range between about 286 mm and about 292 mm, and in some embodiments (e.g., embodiments configured to hold 300 mm workpieces) the diameter is about 288 mm. In some embodiments, the radial distance from an outer edge of a workpiece to the workpiece contact zone **210** can be in a range between about 10 mm and about 12 mm, and in some embodiments, about 12 mm.

(33) The susceptor body can include a workpiece retaining portion **214**. The workpiece retaining portion **214** can include a radially-inward facing circumferential rim. The workpiece retaining portion **214** may be positioned radially outward from the one or more openings **222** and may be configured to prevent radial movement of the workpiece. This can prevent damage to the workpiece and promote higher quality deposition of the workpiece. The workpiece retaining portion **214** can extend to a greater elevation than the workpiece contact zone **210**, to retain a workpiece supported on the workpiece contact zone. The susceptor body **200** can include a circumferential edge facing radially outward of the workpiece retaining portion **214**. In some embodiments, the workpiece retaining portion **214** can have an elevation above the front face **202** in a range between about 0.8 mm and about 1 mm, and in some embodiments the elevation is about 0.9 mm.

(34) The susceptor body **200** may include an inner vacuum region within the inner portion **226** of the front face **202**. The vacuum region can be radially inward of the opening(s) **222**, axial channels(s) (e.g., circumferential purge channel **224**) and/or the workpiece contact zone **210**. The inner vacuum region may include one or more vacuum grooves extending into the front face **202**. For example, a circumferential groove **236** may extend into the front face **202** to form an outer vacuum boundary. One or more radial vacuum grooves can extend into the front face **202**. The inner vacuum region may be split into one or more sections by the one or more radial vacuum grooves **246**. The radial vacuum grooves **246** may form 2, 3, 4, 5, 6, or more sections of the inner vacuum region. The susceptor body **200** may include one or more vacuum apertures **244** that extend into the inner vacuum region of the front face **202**. The vacuum apertures **244** may be in fluid communication with a vacuum. When the workpiece is on the front face **202**, the vacuum may be applied via the vacuum apertures **244** and apply negative pressure to the workpiece along the radial vacuum grooves **246**, the circumferential groove **236**, and/or other regions of the inner vacuum region. The susceptor body **200** may include embossed and/or raised features in the inner vacuum region, such as the raised features **232** shown in FIG. 3. The raised features **232** may provide a small separation between the workpiece and a portion of the front face **202** (e.g., the inner vacuum region). This separation may improve the functionality and efficacy of the applied vacuum. The raised features **232** may help reduce sticking of the workpiece to the susceptor body **200**, and/or may reduce direct contact with the backside of the workpiece, which in turn can reduce contamination or potential workpiece damage. The raised features **232** may also improve the uniformity of heat conduction to the workpiece **206**.

(35) One or more lift pin holes **228** may be included that allow for lift pins to extend through the body **200**, and allow a workpiece (e.g., a wafer) to be placed on the front face **202** and/or removed therefrom. The lift pin holes **228** may be disposed radially inward of the circumferential groove **236** or radially outward therefrom. In some embodiments, there are three lift pin holes **228**, but another number is possible.

(36) FIG. 4A shows a schematic cross-sectional side view of a portion of the susceptor body **200** described above with respect to FIG. 3. For example, FIG. 4A shows an axial purge channel **252**

and opening **222** disposed within the susceptor body **200**, between the front face **202** and the back face **204**. In some embodiments, each of the axial purge channels **252** may be in fluid communication via the circumferential purge channel **224** (FIG. 3).

(37) The workpiece **206** is shown in contact with the workpiece contact zone **210** of the front face **202**. As shown, the workpiece contact zone **210** can be at a greater elevation than the outer portion **220** of the face **202**. In some designs a fluid gap **270** can be formed between the workpiece **206** and the front face **202** radially outward of the workpiece contact zone **210**. The embodiment shown can allow for purge fluid to flow through the axial purge channel **252** and opening **222**, within gap **270** along the backside of the workpiece **206**, and around the outer edge of workpiece **206**. This configuration can reduce or prevent gas used for deposition on the front side of the workpiece **206** from depositing material on the edge or backside of the workpiece **206**. In some embodiments, purge can flow through a circumferential purge channel, such as channel **224** shown in FIG. 3. In some embodiments, the fluid gap **270** can be in a range between about 0.1 mm and about 0.18 mm, and in some embodiments the fluid gap **270** is about 0.15 mm.

(38) FIG. 4A also shows further detail of an embodiment wherein the workpiece retaining portion **214** is disposed at a greater elevation than the workpiece contact zone **210** as described above with respect to FIG. 3.

(39) FIG. 4B shows a perspective cross-sectional view of an example susceptor body **200** with a pedestal **218**. The pedestal **218** can be used to support susceptor body **200**, for example, by supporting via attachment to the back face **204**. The pedestal **218** may additionally or alternatively include one or more channels therein. For example, one or more pedestal purge channels **260** can be disposed within the pedestal **218**. The one or more pedestal purge channels **260** may extend longitudinally through the pedestal **218** and may be configured to be in fluid communication with at least one of the plurality of the openings **222**. For example, the susceptor body **202** can include two axial purge channels **252** as described elsewhere herein. The susceptor body **200** may include one or more radial purge channels **248** positioned between the front face **202** and the back face **204**. The one or more radial purge channels **248** may extend from and/or be in fluid communication with at least one of the axial purge channels **252**, and can provide fluid communication between the axial channels **252** and the longitudinal pedestal channels **260**. In some embodiments, a circumferential groove, shown as a circumferential purge channel **256**, can be in fluid communication with (e.g., provide fluid communication between) the radial channels **248** and the axial channels **252**. In some embodiments, the pedestal purge channels **260** can have a diameter in a range of between about 3 mm and about 5 mm, and in some embodiments is about 4 mm. The radial purge channels **248** can have a diameter in a range of between about 2 mm and about 4 mm, and in some embodiments is about 3 mm.

(40) FIG. 4C shows an example susceptor body **200** that includes axial purge channels **252** that connect the front face **202** with the back face **204**. As best shown in the detail view, in some embodiments the axial purge channels **252** may pass completely through the body **200**, from and through the front face **202** to the back face **204**, and through the back face **204**. As also described above with respect to FIG. 3, the susceptor body **200** can include lift pin holes **228**. As shown, the pedestal **218** may be attached to or include other features, such as an anchoring assembly to provide a secure attachment to a lifting mechanism, fluidics, and/or other features.

(41) FIG. 5 shows a cross-sectional view of an example susceptor body **200**. In some embodiments, the view shown in FIG. 5 is a view of a cross section along a plane different from (e.g., orthogonal to) the view shown in FIG. 4C, and shows many similar vacuum features discussed above and shown in FIG. 3. As shown here, a susceptor body **200** can include one or more pedestal vacuum channels **266** that are in fluid communication with one or more radial vacuum grooves **246**. The circumferential groove **236** is also shown. The pedestal vacuum channels **266** may be disposed radially near an edge of the pedestal **218**. The pedestal vacuum channels **266** may be longitudinal (to form a longitudinal vacuum channel) and may extend through the pedestal **218**. In some

embodiments, the vacuum channels **266** are configured to be in fluid communication with one or more of the vacuum apertures **244**. Additionally or alternatively, the pedestal vacuum channels **266** may be in fluid communication with at least one of the plurality of radial vacuum grooves **246**. The radial vacuum grooves **246** may be in fluid communication with the vacuum apertures **244** and/or the circumferential groove **236**. In some embodiments, the vacuum holes **244** can have a separation distance from one another in a range of between about 43 mm and about 52 mm, and in some embodiments is about 51.5 mm.

(42) FIG. **6** shows a perspective of a cross-section of an example susceptor body **200**. The cross-section may be along a plane different from that shown in FIGS. **4A** and **4B**. As shown, the circumferential purge channel **256** may be disposed throughout the susceptor body **200**. This arrangement of the circumferential purge channel **256** can allow for greater access to and penetration of any purge gas throughout the susceptor body **200** (e.g., between the front face **202** and the back face **204**).

(43) A thermocouple channel, such as a radial thermocouple channel **262** is also shown. The thermocouple channels herein can be configured to include a thermocouple extending therein. The susceptor body **200** may include a thermocouple disposed at least partially within the thermocouple channel **262**. The susceptor body **200** may include one or more radial thermocouple channels **262**. The radial thermocouple channel **262** may be positioned between the front face **202** and the back face **204**.

(44) The pedestal **218** may include a longitudinal thermocouple channel **264** extending longitudinally therethrough that is configured to receive a thermocouple therein. The longitudinal thermocouple channel **264** may be in fluid communication with the thermocouple channel **262**. The thermocouple may be configured to measure a temperature at various points extending radially through the thermocouple channel **262**. Such a long thermocouple channel **262** can allow access to more precise temperature information throughout a greater portion of the susceptor body **200** and/or allow for modifications to the deposition process as needed.

(45) FIG. **7** shows a cross-sectional top view of an example susceptor body **200** between the front face **202** and the back face **204**. As shown, the susceptor body **200** can include a plurality of radial purge channels **248**. The radial purge channels **248** may be in fluid communication with corresponding pedestal purge channels **260**, as shown. One or more groupings of radial purge channels **248** may be associated with corresponding pedestal purge channels **260**. Each of the radial purge channels **248** within a grouping may be radially offset from each other. Additionally or alternatively, radial purge channels **248** between groupings may be offset from each other. For example, a radial offset (e.g., within grouping, between grouping) may be between about 5 degrees and 140 degrees. For example, a radial offset may be about between 50 degrees and 70 degrees. In some embodiments, the radial offset may be approximately equal between neighboring radial purge channels **248** within the same grouping. Additionally or alternatively, the offset may be approximately equal among all neighboring radial purge channels **248**. The thermocouple channel **262** may be disposed approximately in the same plane as one or more of the radial purge channels **248**, as shown.

(46) FIG. **8** shows an example method **300** for purging a workpiece susceptor, such as embodiments of susceptors that include the susceptor bodies **200** described with reference to FIGS. **3-7**, using a processing apparatus, such as apparatus **100**, and in some embodiments, a controller, such as controller **113**, described with reference to FIGS. **2-2**. The method **300** may include loading a workpiece onto a workpiece contact zone on a front face of a susceptor body such that an outer edge of the workpiece is disposed radially outward from the workpiece contact zone, as shown at block **304**. The method **300** can include positioning the workpiece in a processing configuration such that the front face of the susceptor body is in fluid communication with a reaction chamber, as shown in block **308**. The positioning may include positioning a back face of the susceptor body so that it is in fluid communication with a loading chamber. As shown in block **312**, the method **300**

can include providing a first pressure within the reaction chamber. At block **316**, the method **300** can include purging a backside of the outer edge of the workpiece by flowing purge gas at a second pressure from a plurality of channels within the susceptor. The purge gas may be passed to a gap radially outward of the workpiece contact zone and axially between the front face of the susceptor plate and the workpiece to the reaction chamber. In some embodiments, the second pressure is greater than the first pressure. In some embodiments, the second pressure is configured to be higher than the first pressure within the loading chamber.

(47) In some embodiments, the workpiece contact zone includes a circumferential rib extending from the front face. In some embodiments, loading the workpiece onto the workpiece contact zone can include loading the workpiece radially inward of a workpiece retaining portion positioned radially outward from the workpiece contact zone. The workpiece retaining portion may be configured to prevent radial movement of the workpiece, such as disclosed herein. Purging the backside of the outer edge of the workpiece may include flowing purge gas through a plurality of axial channels disposed within the susceptor body. The axial channels can extend through the workpiece susceptor body and the back face of the susceptor body.

(48) In some embodiments, purging the backside of the outer edge of the workpiece includes flowing purge gas through a plurality of openings extending into an outer portion of the front face, each of the openings disposed radially outward of the workpiece contact zone of the susceptor body. One more of the openings may be disposed in fluid communication with corresponding channels of the plurality of axial channels. In some embodiments, purging the backside of the outer edge of the workpiece can include flowing purge gas through a plurality of radial channels that are positioned between the front face and the back face. The radial channels can extend from and/or be in fluid communication with at least one of the plurality of axial channels.

(49) The method **300** may include applying a vacuum to the backside of the workpiece via a plurality of apertures extending into the inner portion of the front face. Applying the vacuum to the backside of the workpiece can include applying the vacuum to at least one longitudinal vacuum channel extending through a pedestal that is configured to support the back face. The at least one longitudinal vacuum channel may be in fluid communication with at least one of the plurality of the apertures. The features of one or more aspects of the method **300** can include functionality and features of the susceptor body **200** disclosed above with respect to FIGS. 1-7.

Illustrative Examples

(50) Various examples are provided below.

(51) In a 1st example, a workpiece susceptor body comprising: a front face configured to support a workpiece; a back face opposite the front face; a workpiece contact zone at least partially forming a support boundary around an inner portion of the front face, the workpiece contact zone configured to be disposed radially inward of an outer edge of a workpiece positioned on the front face in a processing configuration; one or more axial channels disposed within the susceptor body, the axial channels connecting to one or more openings extending into an outer portion of the front face, each of the openings disposed radially outward of the workpiece contact zone of the susceptor body; wherein the workpiece contact zone is at a greater elevation than the outer portion of the face to form a gap radially outward of the workpiece contact zone and axially between the face of the susceptor body and the workpiece.

(52) In a 2nd example, the workpiece susceptor body of example 1, wherein the workpiece contact zone comprises a circumferential rib extending from the front face.

(53) In a 3rd example, the workpiece susceptor body of any of examples 1-2, further comprising the workpiece.

(54) In a 4th example, the workpiece susceptor body of any of examples 1-3, further comprising a workpiece retaining portion positioned radially outward from the openings, the workpiece retaining portion configured to prevent radial movement of the workpiece.

(55) In a 5th example, the workpiece susceptor body of example 4, wherein the workpiece retaining

portion is disposed at a greater elevation than the workpiece contact zone.

(56) In a 6th example, the workpiece susceptor body of any of examples 1-5, wherein the axial channels extend through the workpiece susceptor body and the back face.

(57) In a 7th example, the workpiece susceptor body of any of examples 1-6, further comprising a plurality of radial channels positioned between the front face and the back face, and extending from and in fluid communication with at least one of the plurality of axial channels.

(58) In an 8th example, the workpiece susceptor body of example 7, further comprising a pedestal configured to support the back face, and at least one longitudinal purge channel extending through the pedestal and configured to be in fluid communication with at least one of the plurality of radial channels.

(59) In a 9th example, the workpiece susceptor body of example 8, further comprising one or more apertures extending into the inner portion of the front face, the plurality of apertures configured to be in fluid communication with a vacuum.

(60) In a 10th example, the workpiece susceptor body of example 9, further comprising at least one longitudinal vacuum channel extending through the pedestal and configured to be in fluid communication with at least one of the plurality of the apertures.

(61) In a 11th example, the workpiece susceptor body of example 10, further comprising a plurality of radial grooves extending into the front face, each of the plurality of radial grooves in fluid communication with at least one of the plurality of apertures.

(62) In a 12th example, the workpiece susceptor body of example 11, further comprising a circumferential groove in fluid communication with the plurality of radial grooves.

(63) In a 13th example, the workpiece susceptor body of example 12, wherein the circumferential groove forms an inner vacuum region on the front face, further comprising a plurality of protrusions extending from the front face within the inner vacuum region.

(64) In a 14th example, the workpiece susceptor body of any of examples 7-13, wherein the plurality of radial channels comprises a plurality of radial fluid channels, the plurality of radial fluid channels further comprising at least one radial thermocouple channel positioned between the front face and the back face, the radial thermocouple channel configured to receive a thermocouple.

(65) In a 15th example, the workpiece susceptor body of any of examples 8-13, wherein the plurality of radial channels comprises a plurality of radial fluid channels, further comprising at least one radial thermocouple channel positioned between the front face and the back face, further comprising at least one longitudinal thermocouple channel extending through the pedestal, the at least one radial thermocouple channel and longitudinal thermocouple channel configured to receive a thermocouple.

(66) In a 16th example, a method for purging a workpiece susceptor comprises: loading a workpiece onto a workpiece contact zone on a front face of a susceptor body, such that an outer edge of the workpiece is disposed radially outward from the workpiece contact zone; positioning the workpiece in a processing configuration wherein the front face of the susceptor body is in fluid communication with a reaction chamber and wherein a back face of the susceptor body is in fluid communication with a loading chamber; providing a first pressure within the reaction chamber; purging a backside of the outer edge of the workpiece by flowing purge gas at a second pressure from one or more channels within the susceptor body, to a gap radially outward of the workpiece contact zone and axially between the front face of the susceptor plate and the workpiece, to the reaction chamber, wherein the second pressure is greater than the first pressure.

(67) In a 17th example, the method of example 16, wherein the workpiece contact zone comprises a circumferential rib extending from the front face.

(68) In a 18th example, the method of any of examples 16-17, wherein loading the workpiece onto the workpiece contact zone comprises loading the workpiece radially inward of a workpiece retaining portion positioned radially outward from the workpiece contact zone, the workpiece retaining portion configured to prevent radial movement of the workpiece.

(69) In a 19th example, the method of any of examples 16-18, wherein purging the backside of the outer edge of the workpiece comprises flowing purge gas through one or more axial channels disposed within the susceptor body, wherein the axial channels extend through the workpiece susceptor body and the back face.

(70) In a 20th example, the method of example 19, wherein purging the backside of the outer edge of the workpiece further comprises flowing purge gas through one or more openings extending into an outer portion of the front face, each of the openings disposed radially outward of the workpiece contact zone of the susceptor body, each of the openings disposed in fluid communication with at least one of the one or more axial channels.

(71) In a 21st example, the method of examples 20, wherein purging the backside of the outer edge of the workpiece further comprises flowing purge gas through a plurality of radial channels positioned between the front face and the back face, and the radial channels extending from and being in fluid communication with at least one of the one or more axial channels.

(72) In a 22nd example, the method of example 21, further comprising applying a vacuum to the backside of the workpiece via one or more apertures extending into the inner portion of the front face.

(73) In a 23rd example, the method of example 22, wherein applying the vacuum to the backside of the workpiece comprises applying the vacuum to at least one longitudinal vacuum channel extending through a pedestal configured to support the back face, the at least one longitudinal vacuum channel in fluid communication with at least one of the one or more apertures.

(74) In a 24th example, the method of any of examples 16-23, wherein the purge gas comprises an inert gas and a reactive gas.

(75) In a 25th example, the method of example 24, wherein the inert gas comprises argon and the reactive gas comprises hydrogen.

(76) In a 26.sup.th example, the method of any of examples 16-23, wherein purging comprises a first purge step comprising a first purge gas consisting essentially of an inert gas, and a second purge step comprising a second purge gas comprising an inert gas and a reactive gas

(77) In a 27.sup.th example, the method of example 26, wherein the second purge step sequentially follows the first purge step.

(78) Additional Considerations

(79) The present aspects and implementations may be described in terms of functional block components and various processing steps. Such functional blocks may be realized by any number of hardware or software components configured to perform the specified functions and achieve the various results. For example, the present aspects may employ various sensors, detectors, flow control devices, heaters, and the like, which may carry out a variety of functions. In addition, the present aspects and implementations may be practiced in conjunction with any number of processing methods, and the apparatus and systems described may employ any number of processing methods, and the apparatus and systems described are merely examples of applications of the invention.

(80) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect or embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or embodiments. Various aspects of the novel systems, apparatuses, and methods are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Based on the teachings herein one skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the novel systems, apparatuses, and methods disclosed herein, whether implemented independently of, or combined with, any other aspect described. For example, an apparatus may be

implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosures set forth herein. It should be understood that any aspect disclosed herein may be embodied by one or more elements of a claim.

(81) It should also be understood that, unless a term is expressly defined in this patent using the sentence “As used herein, the term ‘.sub._____’ is hereby defined to mean . . .” or a similar sentence, there is no intent to limit the meaning of that term, either expressly or by implication, beyond its plain or ordinary meaning, and such term should not be interpreted to be limited in scope based on any statement made in any section of this patent (other than the language of the claims). To the extent that any term recited in the claims at the end of this patent is referred to in this patent in a manner consistent with a single meaning, that is done for sake of clarity only so as to not confuse the reader, and it is not intended that such claim term be limited, by implication or otherwise, to that single meaning.

(82) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

(83) Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(84) Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount, depending on the desired function or desired result.

(85) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

(86) Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or

any novel combination, of the steps of any method or process so disclosed.

(87) Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

(88) Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products. For example, any of the components for an energy storage system described herein can be provided separately, or integrated together (e.g., packaged together, or attached together) to form an energy storage system.

(89) For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

(90) The headings provided herein, if any, are for convenience only and do not necessarily affect the scope or meaning of the devices and methods disclosed herein.

(91) The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

Claims

1. A method for purging a workpiece susceptor, the method comprising: loading a workpiece onto a workpiece contact zone on a front face of a susceptor body, such that an outer edge of the workpiece is disposed radially outward from the workpiece contact zone, wherein the workpiece contact zone is substantially annular, and wherein loading the workpiece onto the workpiece contact zone comprises loading the workpiece radially inward of a workpiece retaining portion of the susceptor body positioned radially outward from the workpiece contact zone, the workpiece retaining portion configured to retain the workpiece; positioning the workpiece in a processing configuration wherein the front face of the susceptor body is in fluid communication with a

reaction chamber and wherein a back face of the susceptor body is in fluid communication with a loading chamber; providing a first pressure within the reaction chamber; and purging a backside of the outer edge of the workpiece by flowing purge gas at a second pressure from one or more channels within the susceptor body, to a gap radially outward of the workpiece contact zone and axially between the front face of the susceptor plate and the workpiece, to the reaction chamber, wherein the second pressure is greater than the first pressure.

2. The method of claim 1, wherein the one or more channels comprises one or more axial channels, wherein purging the backside of the outer edge of the workpiece comprises flowing the purge gas through the one or more axial channels disposed within the susceptor body, wherein the one or more axial channels extend through the workpiece susceptor body and the back face.

3. The method of claim 1, wherein the purge gas comprises an inert gas and a reactive gas.

4. The method of claim 3, wherein the inert gas comprises argon and the reactive gas comprises hydrogen.

5. The method of claim 1, wherein loading the workpiece onto the workpiece contact zone comprises loading the workpiece at a greater elevation than an outer portion of the front face to form the gap radially outward of the workpiece contact zone and axially between the front face of the susceptor body and the workpiece.

6. The method of claim 1, wherein the workpiece retaining portion is positioned radially outward from openings within the susceptor that are in fluid communication with the one or more channels.

7. The method of claim 6, wherein loading the workpiece onto the workpiece contact zone comprises supporting the workpiece on the workpiece contact zone at a lower elevation than the workpiece retaining portion.

8. The method of claim 1, wherein the one or more channels comprises a plurality of radial channels and a plurality of axial channels, wherein purging the backside of the outer edge of the workpiece comprises flowing the purge gas through the plurality of radial channels positioned between the front face and the back face, and wherein the plurality of radial channels extend from and are in fluid communication with at least one of the plurality of axial channels.

9. The method of claim 8, further comprising evacuating gas between the workpiece and the susceptor via at least one longitudinal vacuum channel extending through a pedestal configured to support the back face.

10. The method of claim 9, wherein evacuating the gas between the workpiece and the susceptor comprises evacuating the gas via one or more apertures extending into an inner portion of the front face.

11. The method of claim 1, further comprising receiving a thermal reading via a thermocouple disposed within at least one radial thermocouple channel positioned between the front face and the back face.
